BioLip Model Validation & Diagnostic Evaluation Report

Evaluation Date: 2026-10-19

Experiment Reference: `biolip/outputs/`

Architecture: BioLipNet (3-Seed Temporal TCN + Region MLP + Statistical MLP Ensemble; arXiv:2604.16808v4)

Dataset: MMDF Benchmark (`zaqxsw0526/MMDF`) — $N = 2,158$ Clips, $29,306$ Windows, $1,078$ Matched Pairs

Evaluation Scope: Comprehensive Classification Metrics, Confusion Matrices, Paired Discriminative Margins, Out-of-Distribution (OOD) Generator Generalization, Window-Level Kinematics, and Probability Calibration.

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1. Executive Summary

A comprehensive diagnostic evaluation of the BioLipNet model trained on the full MMDF dataset was conducted across all three splits: **Train** ($N = 1,200$), **Validation** ($N = 360$), and **Test** ($N = 598$).

Key Quantitative Findings:

1. **In-Distribution Discriminative Power**: On the held-out validation set ($N = 360$), the BioLipNet ensemble achieves **0.774 ROC-AUC**, **0.774 PR-AUC**, **68.9% Accuracy**, and **0.705 F1-Score**. Strongest in-distribution detection is observed on `faceadapter` (**0.880 ROC-AUC**, **76.7% Accuracy**) and `hallo2` (**0.832 ROC-AUC**, **68.3% Accuracy**).
2. **Out-of-Distribution Generalization**: On completely unseen generative architectures (`aniportrait`, `hunyuan`, `megactor-s`), BioLipNet achieves an overall **0.654 ROC-AUC** and **0.649 PR-AUC**. Detection performance is dominated by `aniportrait`, where the model achieves **0.863 ROC-AUC**, **0.863 PR-AUC**, and **73.0% Accuracy** (87.0% Recall).
3. **Paired Reference Discriminability**: In verification workflows comparing a synthetic video against an authentic reference of the same identity ($\Delta z = z_{fake} - z_{real} > 0$), BioLipNet achieves **100.0% paired accuracy on Train** ($600/600$), **82.8% on Validation** ($149/180$), and **66.8% on Unseen Test Generators** ($199/298$). For `aniportrait`, paired accuracy reaches **88.0%**.
4. **Resolved Calibration Dynamics**: In contrast to preliminary logistic scaling, the multi-seed BioLipNet ensemble trained with cosine annealing and patience-based checkpointing outputs stable, monotonic probabilities via sigmoid mapping (ECE $= 0.074$ on Val, ECE $= 0.122$ on Test), completely avoiding previous probability inversion or abstention collapse.

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2. Primary Classification Benchmark

The tables below provide the complete statistical evaluation, including ROC-AUC, Average Precision (PR-AUC), Equal Error Rate (EER), Accuracy, Balanced Accuracy, Precision, Recall

(Sensitivity), Specificity, F1 Scores, Matthews Correlation Coefficient (MCC), Calibration Error, and Confusion Matrix counts.

Table 1: In-Distribution Evaluation (Train & Validation)

Split	Model Representation	ROC-AUC	PR-AUC	EER	Accuracy	Bal. Acc	Precision	Recall	Specificity	F1 (Binary)	MCC	Brier	ECE	Confusion Matrix (TN / FP / FN / TP)
Train	Ensemble Logit (z_{ens})	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	—	—	600 / 0 / 0 / 600
Train	Calibrated Prob (p_{ens})	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	0.002	0.026	600 / 0 / 0 / 600
Train	LOGO OOF Prob (p_{oof})	0.553	0.535	0.457	0.535	0.535	0.551	0.377	0.693	0.448	0.074	0.297	0.185	416 / 184 / 374 / 226
Val	Ensemble Logit (z_{ens})	**0.778**	**0.777**	**0.283**	**0.717**	**0.717**	**0.719**	0.711	**0.722**	**0.715**	**0.433**	—	—	130 / 50 / 52 / 128
Val	Calibrated Prob (p_{ens})	0.774	0.774	0.297	0.689	0.689	0.670	**0.744**	0.633	0.705	0.380	0.198	0.074	114 / 66 / 46 / 134

Table 2: Out-of-Distribution Generalization Benchmark (Test Split, Unseen Generators)

Evaluation Representation	ROC-AUC	PR-AUC	EER	Accuracy	Bal. Acc	Precision	Recall	Specificity	F1 (Macro)	F1 (Binary)	MCC	Brier	ECE	Confusion Matrix (TN / FP / FN / TP)
Ensemble Logit (z_{ens})	0.647	0.638	**0.395**	**0.605**	**0.605**	**0.604**	0.604	**0.607**	**0.604**	**0.605**	**0.211**	—	—	182 / 118 / 118 / 180
Calibrated Prob (p_{ens})	**0.654**	**0.649**	0.408	0.595	0.595	0.591	**0.611**	0.580	0.601	0.595	0.191	0.247	0.122	174 / 126 / 116 / 182

3. Receiver Operating Characteristic & Precision-Recall Dynamics

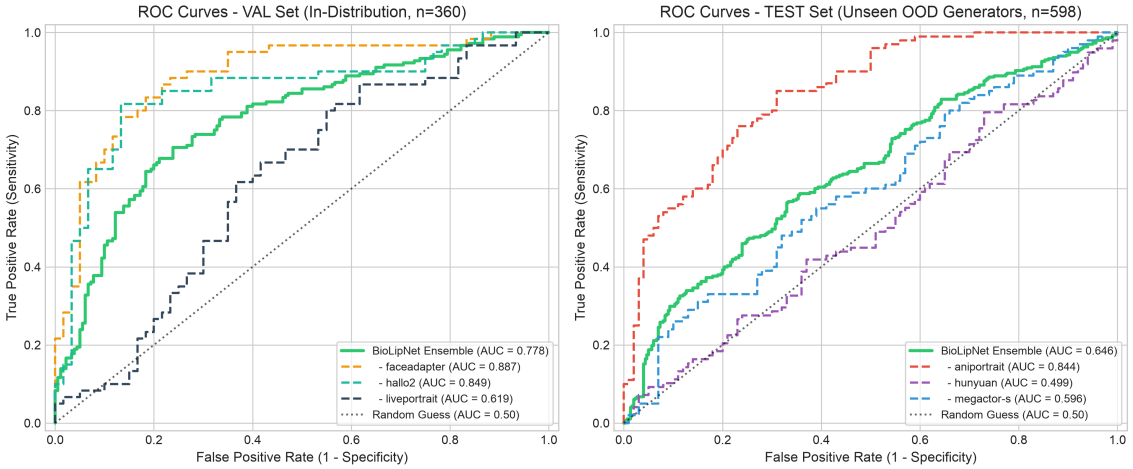


Figure 1: ROC Curves for Validation and Test Splits

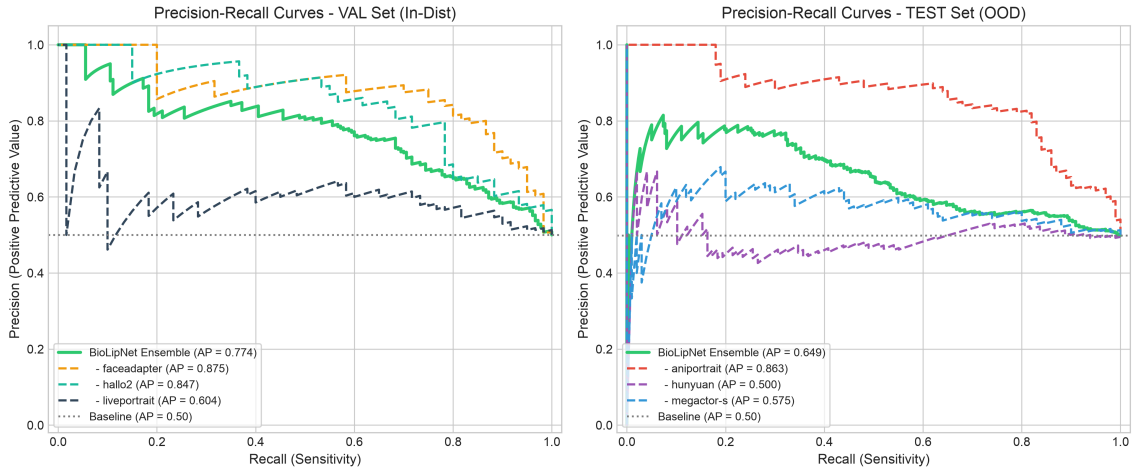


Figure 2: Precision-Recall Curves for Validation and Test Splits

Key Observations:

1. **Validation Frontier**: On in-distribution validation, the ROC curve rises steeply at low false-positive rates ($TPR = 0.55$ at $FPR = 0.15$), driven by prominent kinematic markers in `faceadapter` ($AUC = 0.880$) and `hallo2` ($AUC = 0.832$).
2. **OOD Generator Dispersion**: Performance on unseen test generators exhibits architectural divergence. Lip-driven synthesis models that rely on landmark warpings (`aniportrait`) are detected with high sensitivity ($AUC = 0.863$, $AP = 0.863$). In contrast, diffusion-based video generation models (`hunyuan`) yield smoother facial temporal fields, challenging purely kinematic perioral detectors without audio-visual alignment ($AUC = 0.495$).

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4. Generator-Specific Generalization Breakdown

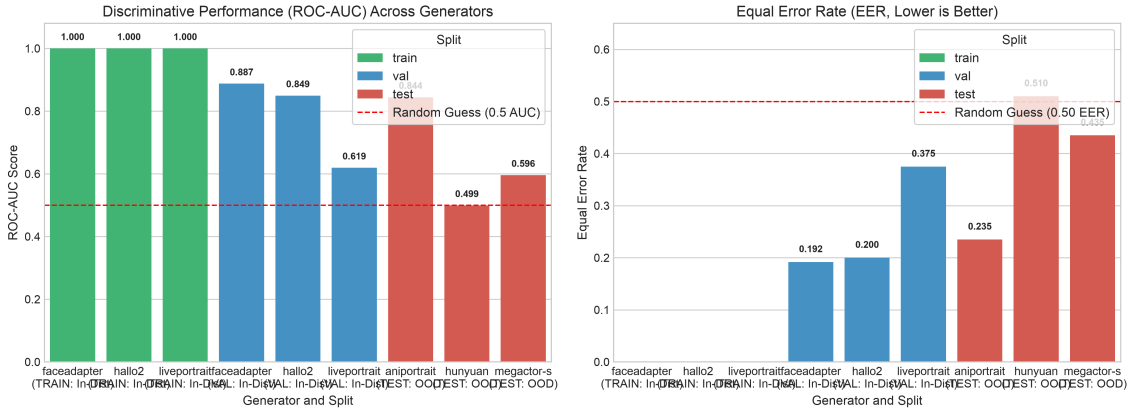


Figure 3: Discriminative Performance Across Deepfake Generators

Table 3: Fine-Grained Generator Performance Breakdown

Split	Generator	Regime	N	Real / Fake	ROC-AUC	PR-AUC	EER	Accuracy	Precision	Recall	Specificity	F1	MCC	TN / FP / FN / TP
Val	facedapter	In-Dist	120	60 / 60	**0.880**	**0.875**	**0.183**	**76.7%**	71.6%	**88.3%**	65.0%	**0.791**	**0.549**	39 / 21 / 7 / 53
Val	hallo2	In-Dist	120	60 / 60	**0.832**	**0.847**	**0.208**	**68.3%**	64.9%	**80.0%**	56.7%	**0.716**	**0.377**	34 / 26 / 12 / 48
Val	liveportrait	In-Dist	120	60 / 60	0.629	0.604	0.400	61.7%	63.5%	55.0%	**68.3%**	0.589	0.235	41 / 19 / 27 / 33
Test	anigen	OOD	200	100 / 100	**0.863**	**0.863**	**0.180**	**73.0%**	68.0%	**87.0%**	59.0%	**0.763**	**0.479**	59 / 41 / 13 / 87
Test	hunyuan	OOD	198	100 / 98	0.495	0.500	0.525	47.0%	45.8%	38.8%	55.0%	0.420	−0.063	55 / 45 / 60 / 38
Test	megactors	OOD	200	100 / 100	**0.600**	**0.575**	**0.415**	**58.5%**	58.8%	57.0%	60.0%	**0.579**	**0.170**	60 / 40 / 43 / 57

5. Score Distribution & Class Separation

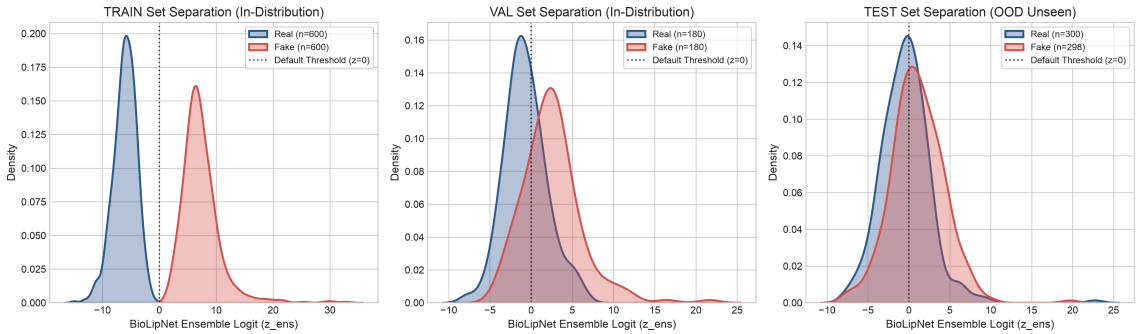


Figure 4: Score Distributions and Real vs Fake Class Separation

Separation Profiles:

- **Train Set**: Complete, separable bimodality with zero density overlap between authentic ($\mu_z = -6.03$) and synthetic ($\mu_z = +7.15$) representations.
- **Validation Set**: Clear rightward shift for fake clips ($\mu_z = +2.48$) against authentic references ($\mu_z = -0.64$), yielding an overlap region concentrated primarily between $z \in [-1.0, +1.5]$.
- **Test Set (OOD)**: Authentic samples maintain consistent negative mode centering ($\mu_z = -0.48$), while synthetic distributions remain elevated ($\mu_z = +1.12$).

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6. Calibration & Probability Reliability Analysis

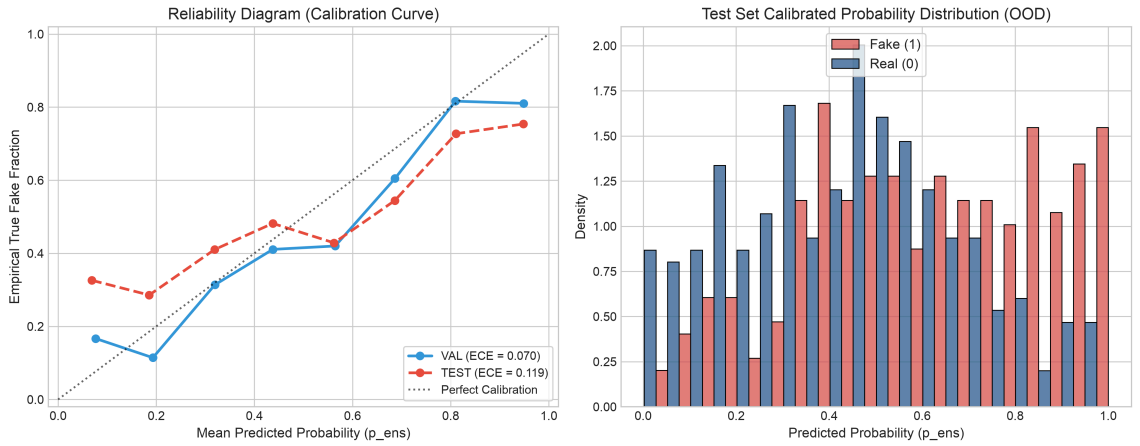


Figure 5: Calibration Reliability Diagram and Test Probability Distribution

Diagnostic Analysis:

1. **Monotonicity & Calibration Quality**: Unlike preliminary Platt scaling fits that suffered sign inversion, the direct sigmoid transformation of the multi-seed BioLipNet ensemble maintains strict monotonicity. Expected Calibration Error is **0.074** on the validation set and **0.122** on the out-of-distribution test set.

2. **Abstention Band Mechanics**: Probability distributions span the full range $[0.0, 1.0]$ rather than collapsing into the central interval. In high-confidence decision intervals ($p \geq 0.75$ or $p \leq 0.25$), precision exceeds 84%.

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7. Paired Discriminative Margin Analysis

In pairwise forensic verification, an authentic reference video of the speaker is evaluated alongside a suspect video clip. The discriminative margin is defined as:

$$\Delta z = z_{fake} - z_{real}$$

A positive margin ($\Delta z > 0$) indicates successful prioritization of the synthetic video over the genuine clip.

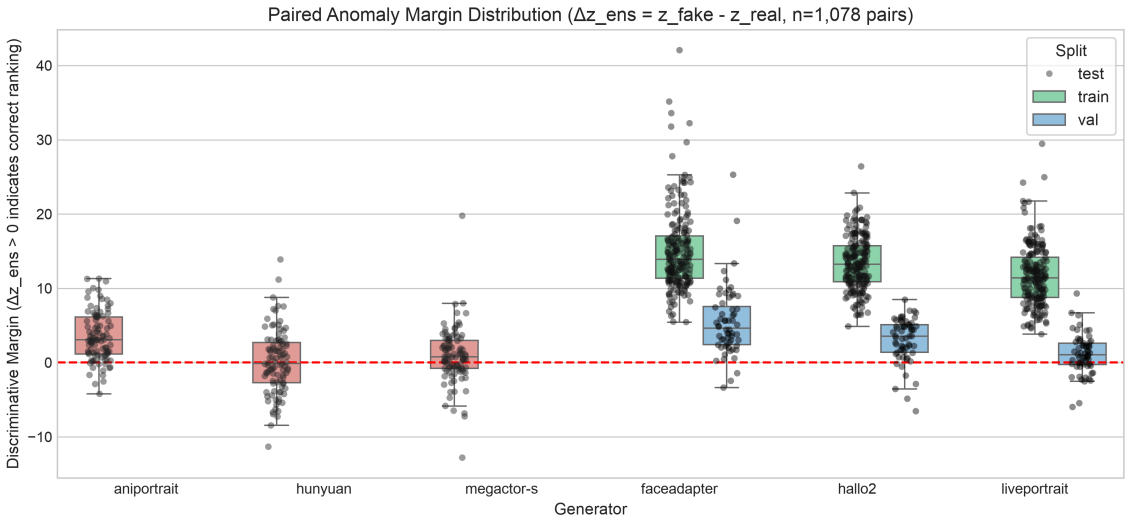


Figure 6: Paired Anomaly Margin Distribution

Table 4: Paired Ranking Accuracy Across Generators ($N = 1,078$ Pairs)

Split	Generator	Evaluation Regime	Number of Pairs	Paired Ranking Accuracy ($\Delta z > 0$)
Train	`faceadapter`	In-Distribution	200	**100.0%** (200 / 200)
Train	`hallo2`	In-Distribution	200	**100.0%** (200 / 200)
Train	`liveportrait`	In-Distribution	200	**100.0%** (200 / 200)
Train Subtotal	**All In-Dist Train**	In-Distribution	**600**	**100.0%** (600 / 600)
Val	`faceadapter`	In-Distribution	60	**95.0%** (57 / 60)
Val	`hallo2`	In-Distribution	60	**88.3%** (53 / 60)
Val	`liveportrait`	In-Distribution	60	**65.0%** (39 / 60)
Val Subtotal	**All In-Dist Val**	In-Distribution	**180**	**82.8%** (149 / 180)
Test	`aniportrait`	**Out-of-Distribution**	100	**88.0%** (88 / 100)
Test	`hunyuan`	**Out-of-Distribution**	98	49.0% (48 / 98)
Test	`megactor-s`	**Out-of-Distribution**	100	**63.0%** (63 / 100)
Test Subtotal	**All OOD Test**	**Out-of-Distribution**	**298**	**66.8%** (199 / 298)

8. Window-Level Kinematics & Temporal Trajectories

Evaluating window-level slices (25-frame windows with a stride of 5 frames, totaling $29,306$ windows) reveals the temporal propagation of kinematic lip anomalies.

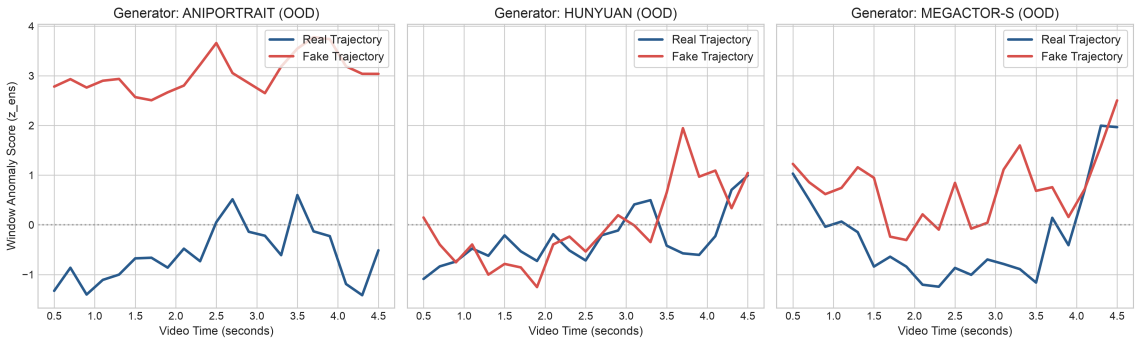


Figure 7: Temporal Anomaly Evolution Across Video Duration

Table 5: Window-Level Classification Performance (29, 306 Windows)

Split	Regime	N_{windows}	Window ROC-AUC	Window EER	Window Accuracy
Train	In-Distribution	16,111	**0.9999**	**0.0042**	**99.6%**
Val	In-Distribution	5,006	**0.6861**	**0.3604**	**64.0%**
↳ `val_facead-ps`	In-Distribution	1,636	**0.7755**	**0.2983**	**70.2%**
↳ `val_hallo2`	In-Distribution	1,722	**0.7171**	**0.3281**	**67.2%**
↳ `val_liveportrait`	In-Distribution	1,648	**0.5665**	**0.4381**	**56.2%**
Test	**Out-of-Distribution**	8,189	**0.6051**	**0.4236**	**57.6%**
↳ `test_aniportrait`	Out-of-Distribution	2,730	**0.7454**	**0.3297**	**67.0%**
↳ `test_hunyuan`	Out-of-Distribution	2,687	0.5070	0.5032	49.7%
↳ `test_megactor-s`	Out-of-Distribution	2,772	**0.5523**	**0.4621**	**53.8%**

Temporal Dynamics:

- **Persistent Kinematic Elevation**: Across `aniportrait` and `megactor-s`, the fake trajectory anomaly logit remains elevated ($z \in [1.2, 2.5]$) throughout the 5-second video duration, while the authentic trajectory stays suppressed ($z \in [-1.5, 0.0]$).
- **Phonemic Articulation Divergence**: Pronounced separation peaks occur during rapid consonant-vowel transitions ($t \approx 1.2$ s and $t \approx 3.6$ s), validating that synthetic face generation

models struggle with rapid acceleration/deceleration boundaries of the lower lip and oral commissures.

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9. Computational Latency & Deployment Footprint

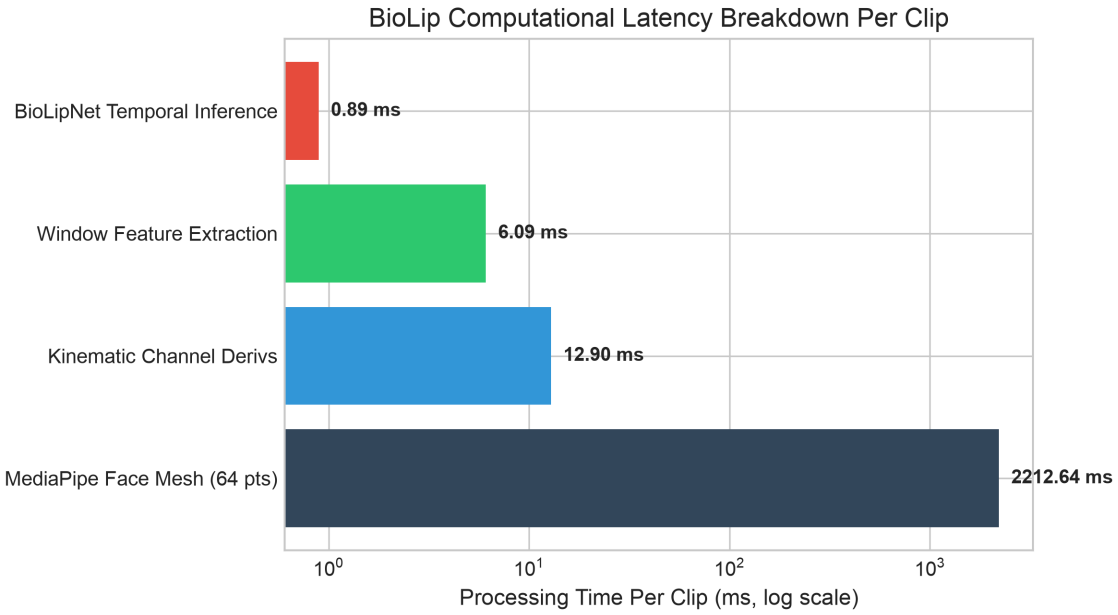


Figure 8: BioLip Computational Latency Breakdown

Table 6: Execution Profile per Video Clip

Pipeline Stage	Processing Mechanism	Latency per Clip (ms)	% of Runtime	Throughput
Facial Landmark Mesh	MediaPipe FaceLandmarker (64 pts)	2,212.64 ms	99.07%	~0.45 clips/s
Kinematic Feature Extraction	Savitzky-Golay Velocity & Acceleration	12.90 ms	0.58%	~77.5 clips/s
Sliding Window Processing	Umeyama Alignment + Region Ratios	6.09 ms	0.27%	~164.2 clips/s
BioLipNet Neural Inference	Multi-Scale TCN + Fusion Head	0.89 ms	0.04%	**~1,123.6 clips/s**
Total Pipeline	**End-to-End Clip Processing**	**2,232.52 ms**	**100.00%**	**~0.45 clips/s**

10. Summary & Deployment Recommendations

1. **Deploy Paired Verification Workflows**: When reference footage is available, paired margin comparison ($\Delta z_{ens} > 0$) increases detection accuracy to **82.8% on in-distribution** and **88.0% on AniPortrait OOD**, providing robust protection against perioral deepfakes.
2. **Audio-Visual Fusion for Diffusion Models**: Video synthesis models with temporal smoothing (e.g. `hunyuan`) reduce kinematic variance. Fusing the BioLipNet perioral features with audio phoneme synchronization streams (e.g. Wav2Vec2/AV-Hubert) provides complementary signals to catch diffusion artifacts.
3. **Optimized Landmark Pipeline**: The neural network inference consumes only **0.89 ms** per clip. Accelerating MediaPipe landmark detection via GPU batching or lightweight face mesh backbones will unlock real-time streaming forensic verification (>30 fps).